# Vincent Ondeng

Geospatial Data Scientist | Spatial Analytics | Earth Observation | Machine Learning | Disease Modelling

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## Profile

Geospatial data scientist with degrees in Geospatial Engineering and Mathematical Sciences. Experience in Earth observation, GIS, multi-source spatial data integration, machine learning on satellite imagery, reproducible geospatial workflows and mathematical modelling of infectious disease. Research interests include spatial epidemiology, environmental determinants of health, climate-related exposures, urban analytics and machine learning for geospatial problems.

## Education

**African Institute for Mathematical Sciences (AIMS)** Kigali, Rwanda

*Master of Science in Mathematical Sciences (infectious disease modelling)* *Sept. 2024 – Oct. 2025*

- **Thesis:** Modeling the Impact of Case Management Practices in Private Drug Retail Outlets on Malaria Treatment Outcomes (supervised at the Center for Epidemiological Modeling and Analysis; details under Research Experience)
- **Relevant coursework:** Statistical Machine Learning, Big Data Analytics, Database and Data Management, Data Structures and Algorithms
- Full scholarship from the Bill and Melinda Gates Foundation

**University of Nairobi** Nairobi, Kenya

*Bachelor of Science in Geospatial Engineering* *Sept. 2018 – Sept. 2023*

- **Thesis:** Mapping Malaria Distribution in Kisumu County, with an interactive web map of the results built in Folium (Python); Best Final Year Project, Institution of Engineers of Kenya
- **Relevant coursework:** GIS, Remote Sensing, Digital Image Processing, Geospatial Data Infrastructure

## Research Experience

**M.Sc. Thesis: Malaria Case Management in Private Drug Retail Outlets** 2025

*AIMS Rwanda and Center for Epidemiological Modeling and Analysis (CEMA)* *Nairobi, Kenya*

- **Question:** how case-management practices in private drug outlets, including failure to test, affect malaria treatment outcomes and continued transmission in Kenya.
- **Data:** the National Malaria Control Programme's 2025 nationwide survey of over 400 private drug retail outlets (outlet audits, mystery-shopper visits, staff interviews), used to set case-management parameters; monthly malaria incidence from the Kenya District Health Information System; CHIRPS rainfall data.
- **Model:** extended SEIR model with treatment-pathway compartments for humans and an SEI structure for mosquitoes, solved in R (deSolve); derived the basic reproduction number, analysed equilibrium stability and showed a forward bifurcation at $R_0 = 1$.
- **Fitting:** five parameters, including the amplitude of rainfall-driven seasonal forcing, estimated by least squares, separately for high-, moderate- and low-burden regions and nationally (own design decision, to capture heterogeneity); model runs parallelised in R.
- **Analysis:** compared four strategies (current practice, universal testing, universal artemether-lumefantrine access, recommended test-and-treat) and ran a global sensitivity analysis with first-order Sobol indices (Saltelli sampling).
- **Results:** test-and-treat improved national recovery rates by about 16.5% and prevented more than 835,000 healthcare re-entries in 2025–2026, with recovery gains up to 24.5% in low- and moderate-burden regions; parameters involving the correct drug explained about 78% of the variance in recovery improvement, testing alone about 8%.
- Sole author of the analysis: model design, coding, fitting, simulation and interpretation.

## Professional Experience

**Geospatial Data Scientist (Consultant)** Oct. 2025 – Sept. 2026

*United Nations Human Settlements Programme (UN-Habitat)* *Nairobi, Kenya*

- Improved and automated workflows for computing SDG indicator 11.3.1 (land consumption relative to population growth) from satellite imagery and demographic data across more than 150 cities in the UN-Habitat city portfolio.
- Classified Landsat and Sentinel imagery from 1990 to 2025 using Random Forest, Support Vector Machine and K-means methods within an automated pipeline for quantifying urban growth in cities worldwide.

- Working independently, combined Google Open Buildings, Microsoft building footprints, World Settlement Footprint and OpenStreetMap, choosing the best-covered source for each city after visual checks against satellite imagery, with fallback to the next source, so the workflow no longer had to classify imagery for every city.
- Corrected over-classified built-up area in 10 m and 30 m rasters derived from building footprints, city by city, for example using NDVI to remove vegetated pixels in green cities.
- With the Namibia Statistics Agency, estimated building counts for Windhoek's informal settlements for 2015, 2020 and 2025 from Google Open Buildings and historical imagery in Google Earth Engine: corrected every 2025 footprint, then worked backwards so earlier years matched one verified baseline.
- Identified and labelled housing typology classes from satellite and street-view imagery to train a convolutional neural network for housing stock estimation in African and Southeast Asian cities.
- Carried out quality assurance and validation of geospatial workflows for realistic urban growth estimates and reproducible outputs.

**M.Sc. Research Fellow** May 2025 – Sept. 2025
*Center for Epidemiological Modeling and Analysis (CEMA)* *Nairobi, Kenya*

- Formulated and fitted the deterministic malaria model described under Research Experience.
- Contributed modelling skills to the National Malaria Control Programme modelling workstream, which was building in-house capacity and seeking evidence for WHO-proposed sub-national tailoring.
- Presented the model at the NMCP Malaria Mathematical Modelling Workshop (Naivasha, July 2025) to the NMCP, Swiss TPH, Mount Kenya University and CEMA, and answered substantial pushback by explaining the model's assumptions.

**Research Assistant** Apr. 2024 – Sept. 2024
*African Population and Health Research Center (APHRC)* *Nairobi, Kenya*

- Integrated Sentinel satellite imagery with Kenya National Bureau of Statistics demographic data to produce spatial indicators, including roof type and building density, for the IDEAMAPS informal settlement mapping effort in Nairobi.
- Set up KoboToolbox data collection for field interviewers collecting water, sanitation and hygiene (WASH) data.
- Overlaid settlement boundaries with hazard layers in ArcGIS, Python and R to identify priority locations for water points, toilets and waste disposal.
- Developed proposals on flood modelling and urban heat resilience using hazard exposure, settlement and demographic data.
- Co-facilitated GIS training for 120 Nairobi County environmental officers and 20 APHRC staff, built around participants' own work with hands-on exercises and group work, and wrote the step-by-step training manual used after the workshops.

**Geospatial Analyst** Jun. 2022 – Mar. 2024
*Widacom Technologies Limited* *Nairobi, Kenya*

- Delivered GIS and spatial analysis for government and private-sector clients, translating requirements into spatial specifications and technical outputs.
- Built and maintained spatial data workflows, including structuring, querying and versioning outputs for repeatable multi-client delivery.
- Automated geoprocessing and data preparation in Python (GeoPandas, Shapely): cleaning, CRS harmonisation, spatial joins and export packaging.
- Produced reporting-ready maps, summary tables and technical notes with systematic quality assurance and quality control.

## Projects

**Cloud-based Environmental Indices Platform** | *Python, FastAPI, STAC, Dask, xarray, Docker* 2026 – Present

- API that takes a GeoJSON boundary and returns NDVI, EVI, SAVI and MNDWI trends and time-series charts from Sentinel-2 imagery, with PNG composites for display and georeferenced GeoTIFFs for GIS use.
- Fixed index values by applying per-collection reflectance scaling (EVI and SAVI had been computed on unscaled digital numbers), and added per-pixel cloud and shadow masking with the Sentinel-2 Scene Classification Layer.
- Restricted band retrieval to each index's needs, roughly halving computation time with identical results; parallelised with Dask; containerised the API and Streamlit front end with Docker Compose.

**Graph Neural Network for Informal Settlement Characterisation** | *PyTorch, GeoPandas* 2026 – Present

- Graph neural network on building-footprint graphs characterising informal settlements in Luanda, Angola and Dhaka, Bangladesh, benchmarked against a multilayer perceptron and Random Forest.
- Compared k-nearest neighbour, Delaunay triangulation and distance-threshold graph constructions.

**Geospatial Language Model for Spatial Coding Tasks** | *Python, LoRA, pyproj* 2026 – Present

- Fine-tuning a small open language model with LoRA to return code with a one-line explanation for GeoPandas operations, PostGIS spatial SQL, CRS handling and STAC queries.
- Supplies each layer's CRS type and units through pyproj lookups in the prompt; plans to compare fine-tuning with a tool-augmented approach and a small transformer trained from scratch.

**LLM-Assisted Application Workflow** | *Claude (custom skills), Python, YAML, LaTeX, Git* 2026 – Present

- Designed an LLM-assisted workflow for job and PhD applications: custom Claude skills for CV tailoring, cover letters, application questions and quality checks, all drawing on a single verified profile.
- Single source of truth for key facts (YAML) with a Python checker that flags superseded figures, wrong dates and inconsistent wording before submission; automated LaTeX builds with page-limit enforcement and rendered page previews for visual checks.
- Used on real applications, where the checks caught factual and formatting errors. Built with Claude; designed, directed and tested by me.

## Manuscripts and Presentations

**Ondeng, V.** Graph-based neural network for informal settlement characterisation from building footprints. *Manuscript in preparation.*

**Ondeng, V.** Modeling the Impact of Case Management Practices in Private Drug Retail Outlets on Malaria Treatment Outcomes. Poster, Malaria Mathematical Modelling Workshop, National Malaria Control Programme, Naivasha, Kenya, July 2025.

**Ondeng, V.**, Opanga, S. and Musiva, A. Improving the Analysis of Harmful Algal Blooms Along the Kenyan Coast by Using Machine Learning and Deep Learning Techniques for Accelerated Satellite Image Processing. Research proposal poster, RCMRD International Conference, 2024 (Best Poster Presentation Award).

## Awards, Scholarships and Certificates

**Bill and Melinda Gates Foundation Scholarship**, AIMS Rwanda: full scholarship for M.Sc. study (2024).

**Best Poster Presentation Award**, RCMRD International Conference (2024).

**Best Final Year Project**, Institution of Engineers of Kenya, Geospatial Engineering (2023).

**Certificates:** Introduction to Research Ethics; Software Engineering Fundamentals.

## Technical Skills

**Programming**: Python (pandas, NumPy, GeoPandas, Shapely, GDAL, Rasterio, xarray, Dask, scikit-learn), R (deSolve), Julia, SQL, Apache Spark, Git and GitHub
**Remote sensing**: Landsat and Sentinel imagery, Google Earth Engine, STAC, spectral indices (NDVI, EVI, SAVI, MNDWI), cloud masking, image classification, satellite and street-view image annotation, climate and environmental data (CHIRPS)
**GIS and web mapping**: ArcGIS Pro, QGIS, PostGIS, Folium
**Machine learning**: Random Forest, SVM, K-means, CNNs (TensorFlow/Keras), graph neural networks (PyTorch), LoRA fine-tuning, supervised and unsupervised methods, feature engineering, model validation, digital image processing
**Modelling**: Compartmental (SEIR) models, stability and reproduction-number analysis, least-squares fitting, Sobol global sensitivity analysis
**Urban analytics**: SDG 11.3.1 and urban indicators, informal settlement mapping, urban risk and vulnerability, housing stock and building morphology
**Data collection and infrastructure**: KoboToolbox, FastAPI, Streamlit, Docker Compose
**Communication**: technical documentation, training manuals, report writing, presenting to technical and non-specialist audiences
**Languages**: Kiswahili (native), English (C1)

## References

**Prof. David Siriba** dnsiriba@uonbi.ac.ke
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*Senior Research Fellow, Center for Epidemiological Modeling and Analysis*

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*Postdoctoral Fellow, CEMA (M.Sc. thesis supervisor)*